

MODULE 01

SYLLABUS: Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.

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FUNDAMENTALS OF ETHICS

What is Ethics?

Ethics is the branch of philosophy that deals with questions of right and wrong, good and bad, and how individuals ought to act in various situations. It provides a systematic framework for evaluating human conduct and making moral decisions.

The word *ethics* comes from the Greek word "**ethos**" meaning character or custom. It is essentially the study of **moral principles** that govern a person's or group's behaviour.

Why Ethics Matters in Engineering: Engineers design systems, structures, and technologies that directly affect human life, safety, and the environment. A wrong calculation may be corrected — but an *ethical failure* can cause irreversible harm. Ethics ensures that technical competence is guided by moral responsibility.

Personal Ethics vs. Professional Ethics

Aspect	Personal Ethics	Professional Ethics
Basis	Individual values, upbringing, beliefs	Standards set by profession/institution
Scope	Private life, personal relationships	Workplace, clients, public
Enforcement	Self-regulated — conscience	External — codes, licensing, law
Flexibility	Varies person to person	Standardised within a profession
Accountability	Answerable to oneself	Answerable to profession, public, law

Personal ethics forms the *foundation*. Professional ethics builds the *structure* on top. Without a strong personal ethical base, professional ethical standards collapse under pressure.

Example: A structural engineer in Kerala is asked to certify a building he privately knows is unsafe — built by a relative. Personal ethics says be honest. Professional ethics *mandates* refusal to certify. The building collapses in Thiruvananthapuram in 2021 traced partly to such lapses — where personal loyalty overrode professional duty with fatal results.

Civic Virtue

The disposition of a citizen — and specifically a professional — to actively prioritise the welfare of the broader community alongside personal and professional interests.

Why it matters for engineers: Engineering decisions are made by a few professionals but their consequences fall on thousands who had no say — residents near a factory, commuters on a bridge, patients using a medical device. This creates a *public trust* relationship that demands civic responsibility.

In practice, civic virtue means:

- Reporting safety violations even when institutionally inconvenient
- Ensuring designs consider community impact, not just client requirements
- Not staying silent when public welfare is threatened

Example: Engineers who raised public concerns about the structural safety of the Mullaperiyar Dam in Kerala — despite significant political pressure to remain quiet — were exercising civic virtue. Their willingness to speak out in the public interest, not for personal gain, is a textbook case of civic virtue in engineering.

Respect for Others

In engineering ethics, respect goes beyond courtesy. It means:

- Recognising the **dignity and rights** of all stakeholders — clients, workers, communities, future users
- Making designs **inclusive** — accessible to people with disabilities, sensitive to cultural contexts
- Giving proper **credit** to colleagues and collaborators
- **Listening** to community concerns before executing a project that affects them

Respect is embedded in virtually every formal code of engineering ethics — IEEE, ASCE, Institution of Engineers (India) — because technical decisions made without respect for those affected produce both ethical and practical failures.

Profession and Professionalism

Profession = A specialised occupation requiring advanced knowledge, formal training, licensing, and a duty of service to society. Engineering, medicine, law — all qualify.

Professionalism = The conduct, values, and standards expected of a member of that profession — *consistently*, not just when being observed.

What separates a *professional* from a skilled worker is the **internalised sense of public obligation**.

Component of Professionalism	Meaning
Competence	Staying current; knowing the limits of your knowledge
Integrity	Honest conduct under pressure, not just convenience
Accountability	Owning errors, not deflecting blame
Service orientation	Public and client interest before personal gain

Ingenuity, Diligence, and Responsibility

Three traits that define the ethical character of an engineer — not just the technical one.

Ingenuity — Creative, effective problem solving. Finding better, safer, more efficient solutions beyond the textbook method. *Warning:* Ingenuity without ethics produces exploitative innovation — e.g., designing addictive algorithms that maximise engagement at the cost of user wellbeing.

Diligence — Thorough, careful effort at every stage. Skipping a site inspection or rushing a structural calculation to meet a deadline is not a minor professional lapse — it is a potential cause of death. Diligence is non-negotiable where public safety is involved.

Responsibility — Owning the consequences of your decisions. The ethical engineer does not say "I was just following instructions." Independent professional judgment is expected, and with it comes accountability.

Example: when given a challenging assignment, they use *ingenuity* to think creatively and find unique solutions instead of copying others. They apply *diligence* by consistently putting in effort, researching thoroughly, and completing the task on time without giving up. At the same time, they demonstrate *responsibility* by managing their time properly, meeting deadlines, and being accountable for the quality of their work.

Integrity in Design, Development, and Research

Integrity = Consistent honesty and transparency across all stages of engineering work.

Domain	What Integrity Demands
Design	Accurately represent capabilities and limitations; maintain safety margins; don't exaggerate performance
Development	Report test results accurately; credit all contributors; refuse shortcuts that compromise quality
Research	No fabrication, falsification, or selective reporting of data; acknowledge funding sources and conflicts of interest

Integrity also means recognising the *boundaries of your competence* — and declaring them — rather than proceeding blindly into unfamiliar technical territory.

Example: After Chandrayaan-2's partial failure, ISRO's transparent and accurate reporting of what succeeded and what did not built institutional credibility that directly enabled Chandrayaan-3's success in 2023. That honesty was an act of professional integrity at an organisational level.

Plagiarism

Using another person's work, ideas, data, or intellectual output without proper acknowledgment and presenting it as one's own.

Forms:

- Direct copying of text, code, or diagrams
- Paraphrasing without citation
- Using another's experimental data without credit
- **Self-plagiarism** — resubmitting your own previously published work as new and original

Why it is a professional ethics issue, not just an academic one: If a plagiarised design contains a flaw, the act of plagiarism has also destroyed accountability — it is impossible to trace whose error it was. It corrupts the entire knowledge base that future engineers rely on.

Standard practice: Cite all sources correctly (IEEE/APA format). Obtain permissions where required. Acknowledge all contributors, including unpaid assistants and data providers.

Example: A B.Tech student submits a final-year project report largely copied from a journal paper with minor word changes and no citation. Even if the technical content is valid, this is an ethical violation. In professional life, the same habit — copying a design without understanding it — can lead to catastrophic misapplication in a different context.

A Balanced Outlook on Law — Challenges

Law = the *minimum* enforceable standard of conduct. **Ethics** = what is *genuinely right*, which often demands more than the law requires.

Three important relationships between law and ethics:

Situation	Example
Legal but unethical	Asbestos companies complied with 1970s regulations while knowingly hiding cancer risks from workers
Illegal but ethically defensible	A whistleblower breaches a confidentiality agreement to expose a life-threatening public hazard
Both legal and ethical	The standard — what every professional should aim for

The real challenge: Emerging technologies — AI, autonomous vehicles, genetic engineering, surveillance systems — have *outpaced* existing legal frameworks. No clear law exists yet. In these situations, an engineer cannot ask only "is this legal?" — they must ask "**is this right?**"

Case Study — Bhopal Gas Tragedy (1984): Union Carbide's Bhopal plant operated within Indian industrial regulations of the time. The methyl isocyanate leak that killed over 3,000 people immediately (and tens of thousands over subsequent years) showed that legal compliance was not sufficient. Engineers inside the plant were aware of deteriorating safety systems. The legal framework failed because the ethical framework had already failed first — professionals chose institutional loyalty over public responsibility.

TECHNOLOGY AND THE DIGITAL REVOLUTION

The digital revolution refers to the shift from mechanical and analogue technologies to digital electronics that began in the latter half of the 20th century and continues to reshape every aspect of human life, work, and society.

Understanding this hierarchy is fundamental to digital ethics.

Level	Definition	Example
Data	Raw, unprocessed facts and figures	"37°C", "500 transactions"
Information	Data processed and given context/meaning	"Patient has fever"; "Sales dropped 20%"
Knowledge	Information applied through experience and understanding to make decisions	"This fever pattern suggests dengue; prescribe accordingly"

Why this matters ethically:

- Data collected without consent is a **privacy violation**
- Information shared without context causes **misinterpretation and harm**
- Knowledge derived from biased data produces **biased decisions** — especially dangerous in AI systems used for hiring, lending, or criminal justice

Example: The Aadhaar biometric database collects raw data (fingerprints, iris scans) from over a billion Indians. When processed, it becomes identity information. When linked to banking, healthcare, and subsidies, it becomes actionable knowledge. Each level of this hierarchy carries increasing ethical responsibility — a breach at the data level can compromise every level above it.

Cybertrust and Cybersecurity

Cybertrust = the confidence users, institutions, and societies place in digital systems to function reliably, securely, and honestly.

Cybersecurity = the set of practices, technologies, and ethical obligations that protect digital systems, data, and networks from unauthorised access, damage, or attack.

Why it is an Engineering Ethics Issue?

Engineers who design software systems, networks, or digital infrastructure are morally responsible for the security of those systems. A poorly designed authentication system is not just a technical flaw — it is an ethical failure, because real people's data, money, and safety depend on it.

Key Ethical Obligations in Cybersecurity:

Obligation	Meaning
Confidentiality	User data must not be accessed or shared without authorisation
Integrity	Data must not be altered or corrupted — systems must ensure accuracy
Availability	Systems must be reliably accessible to authorised users
Accountability	Security breaches must be disclosed honestly and promptly

These three — Confidentiality, Integrity, Availability — form the **CIA Triad**, the foundational model of information security.

Example: In 2021, the CoWIN vaccine registration portal in India suffered a data breach exposing the personal data of millions of citizens. This was both a technical failure and an ethical one — the engineers responsible for the system had an obligation to secure sensitive health data. The breach eroded public trust in digital health infrastructure at a critical moment.

Data Collection and Management — Ethical Dimensions

Data collection today is pervasive — smartphones, apps, sensors, CCTV, and online platforms generate enormous volumes of personal data. The ethical engineer must understand not just *how* to collect data, but *whether* and *how* it should be collected.

Key Ethical Principles in Data Management

Informed Consent: Users must know what data is being collected, why, and how it will be used — and must genuinely agree, not be coerced through buried terms and conditions.

Data Minimisation: Collect only what is necessary for the stated purpose. Collecting excess data "just in case" is an ethical violation even if not currently illegal.

Purpose Limitation: Data collected for one purpose must not be repurposed without fresh consent. Health data collected for a wellness app must not be sold to insurance companies.

Data Security: Collected data must be protected against breaches with appropriate technical measures.

Right to Erasure: Individuals should have the ability to request deletion of their data when it is no longer needed.

Example: Facebook's Cambridge Analytica scandal (2018) — data collected from Facebook users (supposedly for academic research) was repurposed for political profiling and used to influence elections in the US and UK. The engineers and data scientists who built these systems faced severe ethical questions: they technically *could* do this — but *should* they have?

High Technologies

High technologies — the internet, mobile networks, AI, GPS, satellite communication — have transformed how people connect, access services, and participate in society.

The Promise and the Problem:

Positive Impact	Negative / Ethical Concern
Remote healthcare (telemedicine)	Digital divide — rural and poor communities left behind
Online education reaching remote areas	Dependence on platforms controlled by corporations
E-governance reducing corruption	Surveillance and tracking of citizens
Financial inclusion via UPI/mobile banking	Cybercrime and fraud targeting vulnerable users

The Digital Divide

The digital divide refers to the gap between those who have meaningful access to digital technologies and those who do not — divided along lines of geography, income, age, gender, and disability.

In Kerala, despite high literacy and mobile penetration, significant gaps exist — elderly citizens struggle with mandatory digital processes (Aadhaar linking, online ration card management), and tribal communities in Wayanad and Attapadi have limited reliable internet access.

Ethical responsibility of engineers: Technology designed without considering accessibility is exclusionary. Universal design principles demand that systems work for the widest possible range of users - including those with disabilities, limited literacy, or slow internet connections.

Managing Conflict

Conflict in a professional setting arises when individuals or groups have incompatible goals, interests, values, or perceptions. Managing conflict ethically means resolving it in a way that is fair, transparent, and preserves professional relationships.

Sources of Conflict in Engineering Practice:

- Technical disagreements (design choices, safety standards)
- Resource allocation disputes (budget, manpower)
- Ethical disagreements (a colleague cutting corners on safety)
- Interpersonal conflicts (credit, recognition, communication failures)

Ethical Approaches to Conflict Resolution:

Approach	Description	When Appropriate
Collaboration	Both parties work toward a mutually acceptable solution	When relationship and outcome both matter
Compromise	Each party gives up something	When both sides have valid points
Avoidance	Temporarily stepping back	Minor, low-stakes issues
Confrontation (assertive)	Directly addressing the issue	Safety or ethical violations — cannot be avoided
Accommodation	Yielding to the other party	When the other party is clearly correct

When the conflict involves public safety — an engineer disagreeing with a supervisor who wants to approve an unsafe design — *avoidance and accommodation are not ethically acceptable*. The engineer has a professional duty to escalate.

Example: During the construction of a flyover in Palakkad, if a junior engineer identifies that the reinforcement used is substandard but the site manager pressures him to stay quiet, the ethical response is not accommodation — it is formal documentation and escalation to higher authority or the relevant regulatory body.

Collective Bargaining

Collective bargaining is the process by which workers, through their trade unions or representative bodies, negotiate with employers regarding wages, working conditions, safety standards, and other employment terms.

Relevance to Engineering Ethics:

Engineers as professionals occupy a dual role — they may be *employees* subject to workplace conditions, and simultaneously *professionals* with public obligations. This creates unique ethical tension:

- As employees, they have rights to fair wages and safe conditions — collective bargaining protects these
- As professionals, they cannot allow labour disputes to compromise public safety

The ethical boundary: A strike by construction engineers that halts a project is a legitimate exercise of labour rights. However, if engineers responsible for maintaining a critical system (dam, power grid, hospital infrastructure) abandon their posts during a crisis, the ethical obligation to public safety overrides the labour right.

Example: The Kerala State Electricity Board engineers' strike in 2022 raised precisely this tension — legitimate grievances about service conditions, but public dependence on uninterrupted power supply created an ethical complexity that required negotiated resolution rather than simple work stoppage.

Confidentiality

Confidentiality is the ethical and often legal obligation to protect information shared in a professional relationship from disclosure to unauthorised parties.

Why Confidentiality is Central to Professional Ethics?

Professional relationships — engineer-client, doctor-patient, lawyer-client — depend on trust. A client shares sensitive technical, financial, or strategic information with an engineer because they trust it will not be disclosed to competitors, the public, or other parties without consent. Violating this trust destroys the foundation of professional practice.

What Must Be Kept Confidential

- Client's proprietary designs, processes, and business information
- Personnel matters within an organisation
- Research data before publication
- Internal safety audit findings (with important exceptions — see below)

The Critical Ethical Limit — When Confidentiality Must Be Broken:

Confidentiality is **not absolute**. When keeping information secret would put the public or individuals at serious risk, the ethical (and sometimes legal) obligation shifts toward *disclosure*.

Situation	Ethical Action
Client's confidential design is merely commercially sensitive	Maintain confidentiality
Confidential internal report reveals a product defect that could injure users	Disclose to appropriate authority
Employer's private financial struggles	Maintain confidentiality
Employer is illegally dumping toxic waste into a river	Whistleblowing is ethically obligatory

The test: If maintaining confidentiality requires becoming complicit in harm to others, confidentiality must yield.

Example: An engineer at a chemical plant near Periyar River (Eloor industrial zone, Ernakulam) discovers internal reports confirming toxic effluent discharge is far above safe limits. Maintaining confidentiality here means protecting corporate interest at the cost of public health — the Periyar pollution crisis is a real case where disclosure was both ethically required and eventually legally pursued.

Role of Confidentiality in Moral Integrity

Confidentiality and moral integrity are interconnected in two directions:

Maintaining confidentiality = moral integrity when it protects legitimate privacy, builds professional trust, and respects the rights of those who shared information.

Breaking confidentiality = moral integrity when silence would make you complicit in serious harm, deception, or injustice.

The person of genuine moral integrity does not use "confidentiality" as a convenient shield to avoid difficult professional decisions. True integrity demands honest judgment about *when* each obligation applies.

Codes of Ethics

A code of ethics is a formal document that sets out the principles, values, and standards of conduct expected of members of a profession or organisation.

Purpose of Codes of Ethics

- Provide clear guidance in ethically ambiguous situations
- Create professional accountability
- Build public trust in the profession
- Protect professionals who act ethically against institutional pressure

Key Engineering Codes of Ethics

Organisation	Key Principles
IEEE (Institute of Electrical and Electronics Engineers)	Public safety first; honesty; avoiding conflicts of interest; improving technical competence
ASCE (American Society of Civil Engineers)	Engineers shall hold public safety paramount; act with integrity; be honest
Institution of Engineers (India)	Competence, integrity, public service, environmental responsibility

Common Principles Across All Codes

1. **Public safety and welfare** come before employer or client interests
2. **Honesty** in all professional communications
3. Act only within areas of **competence**
4. Avoid **conflicts of interest** or disclose them
5. Treat all people with **fairness and respect**
6. Protect the **environment**
7. Continue professional **development and learning**

Limitations of Codes of Ethics

Codes cannot anticipate every situation. They provide a framework, not a formula. Real ethical judgment requires applying principles to novel situations — which is why understanding the *reasoning* behind codes matters more than memorising their text.

Example: The IEEE Code of Ethics requires members to "disclose promptly factors that might endanger the public or the environment." This directly obligates an IEEE-member engineer who discovers a cybersecurity flaw in a critical infrastructure system to report it — even if their employer wants it kept quiet for commercial reasons.

GENDER STUDIES & ETHICS

These foundational concepts are frequently confused. Understanding the distinctions is essential — both for exam answers and professional practice.

Sex

The biological classification of humans as male, female, or intersex, based on chromosomes, hormones, and reproductive anatomy. Sex = *biological*

- Determined at birth (or before) by biological factors
- Binary in conventional understanding — but biology itself shows variation
- **Intersex** conditions (approximately 1.7% of population) — individuals born with reproductive or sexual anatomy that doesn't fit typical male/female definitions

Gender

The socially and culturally constructed roles, behaviours, expressions, and identities associated with being a man, woman, or another gender. Gender = *social and cultural construct*

- Not determined by biology — constructed by society, culture, and history
- Varies across cultures and changes over time
- What is considered "masculine" or "feminine" in Kerala in 2024 differs from what it was in 1950, and differs from norms in Scandinavian countries

Sexuality

A person's pattern of emotional, romantic, and/or sexual attraction to others.

Term	Meaning
Heterosexual	Attracted to the opposite gender
Homosexual (Gay/Lesbian)	Attracted to the same gender
Bisexual	Attracted to both same and other genders
Asexual	Little or no sexual attraction to others
Pansexual	Attracted regardless of gender identity

- Sexuality is **independent** of both sex and gender
- It exists on a spectrum, not as fixed categories

The Key Distinction — Sex vs. Gender vs. Sexuality

	Sex	Gender	Sexuality
Nature	Biological	Social/Cultural	Emotional/Romantic
Assigned by	Biology	Society/Self	Self
Fixed?	Largely, but intersex variations exist	No — fluid and cultural	No — can be fluid

The Gender Spectrum — Beyond the Binary

Traditional thinking presents gender as a strict binary: male or female. Contemporary understanding recognises this as insufficient.

Gender spectrum = the concept that gender exists along a continuum, with many possible identities beyond just "man" or "woman."

Identity	Meaning
Cisgender	Gender identity matches sex assigned at birth
Transgender	Gender identity differs from sex assigned at birth
Non-binary	Does not identify exclusively as man or woman
Genderfluid	Gender identity shifts over time
Agender	Does not identify with any gender
Genderqueer	Rejects conventional gender distinctions

Indian legal context: The Supreme Court of India in **NALSA v. Union of India (2014)** recognised transgender persons as a third gender, affirming their constitutional rights. The **Transgender Persons (Protection of Rights) Act, 2019** further formalised legal recognition — though activists have raised concerns about its implementation gaps.

Kerala context: Kerala has been relatively progressive — the state government launched the **Social Justice Department's transgender welfare schemes** and the **Transgender Policy (2015)**, one of the first state-level transgender policies in India.

Gender Identity and Gender Expression

Gender Identity

A person's deeply held, internal sense of their own gender — how they perceive themselves.

- This is *private* and *self-determined*
- May or may not correspond to biological sex
- Cannot be assigned, corrected, or changed by external forces

Gender Expression

The external presentation of gender — through clothing, behaviour, hairstyle, speech, and mannerisms.

- This is *public* and *visible*
- May or may not match gender identity
- Policed heavily by social norms — with real consequences

	Gender Identity	Gender Expression
Nature	Internal, private	External, visible
Determined by	Self	Chosen, but shaped by social pressure
Others can observe?	No	Yes

Example: A transgender woman may have a female gender identity (internal) but may express herself cautiously in public due to fear of discrimination — showing that gender expression is shaped not just by identity but by social safety.

Gender Stereotypes

Oversimplified, generalised beliefs about the roles, abilities, and characteristics that men and women are expected to have.

Common Gender Stereotypes in Engineering Context:

Stereotype	Reality
"Engineering is a man's field"	Women have contributed fundamentally to engineering and computing throughout history
"Women are better at soft skills, men at technical skills"	No credible evidence supports innate gender-based skill differences
"A woman engineer must be unusually exceptional to succeed"	Men do not face this standard — the bar is unequal
"Men don't need parental leave"	This stereotype harms both men and children

How Stereotypes Cause Real Harm

- **Self-limiting behaviour** — girls internalise "engineering is not for me" early
- **Hiring bias** — resumes with female names receive fewer callbacks in technical roles (documented in multiple studies)
- **Workplace culture** — environments where women must constantly prove themselves create attrition
- **Design bias** — products designed by male-dominated teams may not account for female users

Example: Early voice recognition systems (including early versions of Siri and Google Assistant) performed significantly worse on female voices — because the training data was predominantly male. A gender-diverse engineering team would likely have caught this. The stereotype that "engineers are men" produced a product that literally did not hear women as well.

Gender Disparity and Discrimination

In Education

Despite Kerala's high literacy rate (96.2%) — nearly equal for men and women — gender disparity persists in specific educational streams:

- Girls are steered away from mathematics and science at secondary level through family and social pressure
- Engineering colleges in Kerala show improving female enrollment, but certain branches (mechanical, electrical) remain male-dominated
- Female students face higher pressure to choose "safe" career paths over ambitious technical ones

In Employment

Form of Discrimination	Description
Glass ceiling	Invisible barrier preventing women from reaching senior positions
Pay gap	Women paid less than men for equivalent work
Occupational segregation	Women concentrated in lower-paying sectors and roles
Maternity penalty	Career consequences for women who take maternity leave
Harassment	Sexual harassment creating hostile work environments

India-specific data: Women constitute approximately 43% of STEM graduates in India — one of the higher rates globally — yet only about 14% of the engineering workforce. The gap between education and employment is the critical problem.

Kerala example: ISRO's Thiruvananthapuram centre employs a significant number of women scientists, and the Chandrayaan-3 mission team included notable women in leadership. Yet at the broader institutional level, women in senior ISRO positions remain underrepresented relative to their entry-level numbers — showing that entry is improving but progression is still unequal.

In Everyday Life:

- Unequal distribution of domestic labour — even when both partners work full-time
- Safety constraints on women's mobility (especially at night) limiting professional opportunities
- Social expectations around marriage and caregiving affecting career continuity

History of Women in Science and Technology

Recognising this history is not merely symbolic — it corrects a distorted narrative that has actively discouraged women from technical fields.

Name	Contribution
Ada Lovelace (1815–1852)	First computer programmer — wrote the first algorithm for Charles Babbage's Analytical Engine
Marie Curie (1867–1934)	First person to win two Nobel Prizes (Physics and Chemistry); discovered polonium and radium
Hedy Lamarr (1914–2000)	Invented frequency-hopping spread spectrum — foundational to modern WiFi and Bluetooth
Katherine Johnson	NASA mathematician whose orbital mechanics calculations were essential to Apollo 11 moon landing
Grace Hopper	Developed the first compiler; popularised the idea of machine-independent programming languages
Radia Perlman	Invented the Spanning Tree Protocol — fundamental to how the internet works

Indian context:

Name	Contribution
Janaki Ammal	Pioneering botanist; first Indian woman to receive a DSc; worked on sugarcane genetics
Anna Mani	Physicist and meteorologist; developed over 100 meteorological instruments; worked on solar radiation measurement
Tessy Thomas	"Missile Woman of India" — Project Director for Agni-IV and Agni-V missile systems at DRDO
Ritu Karidhal	Mission Director for Chandrayaan-2; Senior scientist at ISRO Bangalore

Key point for exam: These women did not succeed *despite* being women — they succeeded *while being systematically excluded* by institutions, denied credit, and often erased from official histories. Recognising them is an act of historical correction.

Gendered Technologies and Innovations

Technologies that are designed, developed, or deployed in ways that embed gender biases — either by excluding women from the design process or by reinforcing gender stereotypes through the product itself.

Examples of Gendered Technology

Crash test dummies: Until recently, the standard crash test dummy was modelled on the average male body. Female-bodied people are statistically more likely to be seriously injured in car crashes — in part because safety systems were designed without them in mind.

Medical research bias: For decades, clinical drug trials excluded women (supposedly to avoid hormonal variation complicating results). Drugs were approved based on male physiology data and then prescribed to women — with different and sometimes dangerous effects.

AI hiring tools: Amazon developed an AI recruitment tool that was found to systematically downgrade resumes containing the word "women's" (as in "women's chess club") — because it was trained on historical hiring data from a male-dominated industry.

Smartphone size: The shift to large-screen smartphones was driven by male-dominated design teams. Research shows women (who on average have smaller hands) have more difficulty using large phones one-handed — a basic usability consideration that was missed.

Kitchen technology and domestic appliances: Historically, most "labour-saving" domestic technology was designed to improve women's efficiency in domestic labour — reinforcing the social expectation that domestic work is women's work — rather than redistributing that labour.

The engineering ethics connection: Gendered technology is not just a social justice issue — it is an *engineering quality* issue. A product that does not work equally well for all users is, by definition, a poorly engineered product.

Ethical Values and Practices — Equity, Diversity, and Gender Justice

Equality vs. Equity — A Critical Distinction

	Equality	Equity
Approach	Give everyone the same thing	Give everyone what they need to reach the same outcome
Assumption	Everyone starts from the same position	Recognises that historical disadvantage creates unequal starting points
Result	May perpetuate existing inequality	Aims to produce genuinely fair outcomes

Simple illustration: Giving every engineering student the same scholarship amount is *equality*. Giving larger scholarships to students from economically disadvantaged or historically marginalised communities is *equity* — because they face greater barriers to completing the same degree.

Diversity in Engineering

Diversity = the presence of people with different genders, backgrounds, identities, and perspectives in a professional team.

Why diversity is an engineering imperative, not just a social one:

- Diverse teams catch more design flaws (different users think of different edge cases)
- Diverse teams produce more innovative solutions (different life experiences = different problem framings)
- Products designed by diverse teams serve a wider user base more effectively

Gender Justice

The fair and equitable treatment of all genders — addressing and rectifying historical and ongoing discrimination, not merely achieving formal legal equality.

Gender justice in engineering practice means:

- Actively recruiting and retaining women and transgender engineers
- Designing products and systems that serve all genders equally
- Creating workplace cultures free from harassment and bias
- Ensuring that the *benefits* of technology are distributed equitably across genders

Gender Policy and Empowerment Initiatives

National Level

Policy / Initiative	Key Features
National Policy for Women (2016)	Framework for gender equality across health, education, employment
POSH Act (2013) — Prevention of Sexual Harassment at Workplace	Mandates Internal Complaints Committees in all organisations with 10+ employees
Transgender Persons (Protection of Rights) Act, 2019	Legal recognition, prohibition of discrimination, welfare measures
Beti Bachao Beti Padhao (2015)	Addresses declining sex ratio; promotes girl child education
Maternity Benefit (Amendment) Act, 2017	Extended paid maternity leave from 12 to 26 weeks

Kerala Level

Initiative	Details
Kerala Transgender Policy (2015)	First state-level comprehensive transgender policy in India
She Lodge scheme	Safe accommodation for working women in urban areas
Gender Park, Kozhikode	Asia's largest gender resource centre — research, training, advocacy
Kudumbashree Mission	Women's self-help network — over 4.5 million members; economic empowerment at grassroots level

International Level

Framework	Key Points
CEDAW (1979) — Convention on Elimination of All Forms of Discrimination Against Women	International treaty; India is a signatory
UN SDG 5 — Gender Equality	Part of Sustainable Development Goals; targets include ending discrimination and ensuring full participation
Beijing Platform for Action (1995)	Global framework for women's empowerment across 12 critical areas

Relevance to Engineers

The **POSH Act** directly affects engineering workplaces — every organisation must have an Internal Complaints Committee. Engineers in leadership positions have a legal and ethical responsibility to ensure this is functional, not just formally existing. Engineers designing public systems — transport, housing, healthcare platforms — must conduct **gender impact assessments** to ensure these systems do not disadvantage women or transgender persons.

MODEL QUESTIONS

1. What is civic virtue? How does it relate to professional ethics? *(3 Marks)*
2. What do you mean by 'gendered technology' and provide one specific example of a product whose design or application reflects this concept. *(3 Marks)*
3. You are part of a team designing an app that collects user data for public services. What ethical concerns would you raise regarding data privacy and consent? Suggest safeguards. *(8 Marks)*
4. What is the fundamental difference between 'Equity' and 'Equality' when discussing gender initiatives and resource allocation in education/workplace? *(5 Marks)*
5. Suggest measures to promote gender equity in workplaces. *(3 Marks)*